

Machine Learning Reflection Journal:

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1. Conceptual Breakthroughs: Learning Paradigms

Working through the notebook, I noticed how fundamentally machine learning paradigms differ based on data structure rather than just syntax:

- **Supervised Learning:** I realized supervised learning depends on having an "answer key" during training (X, y). In our wine classification task, mapping 13 chemical features to predict discrete cultivars (class_0, class_1, class_2) helped me see that classification defines boundaries, whereas regression extends this to continuous targets.
- **Unsupervised Learning:** I observed that unsupervised models work strictly on unlabeled feature space (X). Without target labels, the algorithm must discover intrinsic patterns, visual groupings, or compressed feature representations completely on its own.
- **Reinforcement Learning:** I noticed this paradigm shifts focus from static data matching to dynamic decisions, where an agent learns through trial and error using scalar feedback (rewards and penalties).

2. Realizations Across the 6-Step ML Workflow

1. **Data Preparation:** Assembling chemical attributes into feature matrix X and target array y showed me how proper feature alignment prevents errors downstream.
2. **Feature Selection:** Evaluating feature variance and class separability revealed that isolating high-signal attributes directly sharpens model decision boundaries.
3. **Data Splitting:** Splitting the dataset (80% train / 20% test) with stratify= y highlighted how critical it is to preserve uniform class proportions across splits to avoid training bias.
4. **Model Training:** Training a linear baseline (**Logistic Regression**) against a depth-limited tree (**Decision Tree**, max_depth=3) allowed me to compare two completely different model structures side by side.
5. **Model Evaluation:** Testing on unseen data (X_{test}) was an eye-opener: Logistic Regression achieved higher generalization (88.9%, 32/36 correct) than the Decision Tree (83.3%, 30/36 correct).
6. **Model Interpretation:** Inspecting confusion matrices showed me exact failure points—while Cultivar 0 was perfectly classified, there was minor boundary overlap between Cultivars 1 and 2.

3. Engineering Observations & Takeaways

- **Controlling Overfitting:** I noticed that setting `random_state` ensures experiment reproducibility, while capping tree depth (`max_depth=3`) prevents the model from memorizing noise.
- **Data Leakage Risk:** I learned that feature scalers must fit exclusively on training data (X_{train}) before transforming testing data (X_{test}) to prevent test statistics from leaking into model training.
- **Metric Diagnostics:** Looking beyond raw Accuracy, I realized evaluating Precision, Recall, and F1-score across classes exposes weaknesses (such as lower recall on `class_2`) that overall accuracy completely hides.

Hamza Ajmal

Reflective Journal: At the start of this lab, machine learning felt more complicated to me because there were several steps involved in getting from a dataset to a model result. I understood the basic idea of training a model, but I was not completely sure how the different parts of the workflow connected. As I worked through the tasks, I started to understand that choosing the features, preparing the data, training the model, and evaluating the results are all connected. The evaluation is not just a final number; it also helps explain whether the choices made earlier were useful. One concept that became much clearer to me was the difference between supervised, unsupervised, and reinforcement learning. The "answer key" idea helped me understand supervised learning because the correct output is already known during training. For example, in the wine classification task, the class label is the answer the model is learning to predict. In unsupervised learning, that answer key is missing, so the model looks for patterns or groups by itself. Reinforcement learning was different because the model learns through rewards and penalties instead of being given the correct answer for every action. Thinking about the three types this way made more sense to me than just memorizing their definitions. The lab also built on things I had already used in Lab 2. I used pandas and Matplotlib there, so those tools were familiar when I encountered them again in this lab. I was more comfortable working with data and using Data Frames and plots to understand it. What felt newer was taking that data and moving into the actual machine learning workflow, including training models, making a train/test split, and evaluating predictions with accuracy, precision, recall, and F1-score. This helped me see how the data-analysis skills from Lab 2 can become the starting point for a machine learning workflow. The model evaluation part changed the way I think about model performance.

Before this lab, I would have been more likely to look at accuracy and choose the model with the larger number. In this lab, Logistic Regression achieved 88.9% accuracy, or 32 correct predictions out of 36, while the Decision Tree achieved 83.3%, or 30 correct predictions out of 36. Logistic Regression therefore had a difference of only two test samples. Looking at the classification reports showed me that accuracy does not tell the entire story. For example, Logistic Regression had a class_2 recall of 0.70, while the Decision Tree had 0.80. So even though Logistic Regression performed better overall, the Decision Tree performed better at identifying class_2 in that test set. One result from Task 1 was especially interesting to me. I saw the message saying, "Great job! Your feature selection improved the model!" after using the features flavonoids, color intensity, and proline. That model achieved 91.7% accuracy compared with 88.9% for the original model. At first, the improvement looked significant, but when I thought about the test set, I realized that 91.7% was 33 out of 36 samples while 88.9% was 32 out of 36. The improvement was therefore only one additional correct prediction. This was a useful lesson for me because it showed why I should not overstate a result just because the percentage looks noticeably higher. Another challenging part was keeping track of which features I was actually using when interpreting the results. I almost used information from the group-means table for one feature set to explain results from models trained on a different set of features. Catching that made me realize that I need to be more careful when connecting evidence to a model. The information must come from the same experiment or feature set, otherwise the explanation might sound reasonable but not actually explain the result I am discussing. If I did the lab again, I would keep better notes while working and record which feature set and model produced each result. I

would also test more variations instead of relying on one train/test split. I would try cross-validation to see whether the Logistic Regression advantage stays consistent and experiment with the Decision Tree's `max_depth` to see how changing its complexity affects performance. I would also pay more attention to scaling because some features, such as `proline`, have values much larger than features measured in single digits. The biggest change in my thinking is that I now see machine learning as more than simply choosing a model and checking its accuracy. I understand the workflow as a series of connected decisions: understanding the data, selecting useful features, training a model, evaluating different aspects of its performance, and questioning whether the observed differences are large enough to matter. The lab also showed me that a result that looks like an improvement may only be one test sample different, so I need to look at the actual evidence before drawing conclusions

Johnna Matthews

1. Introduction

This week I learned about three types of machine learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. I learned that each type learns in a different way depending on the type of data and the goal of the problem. We also practiced using machine learning with the Wine dataset, which helped me understand how a model can learn from data and make predictions.

2. Description of Experience or Topic

Background Information

Machine learning allows computers to learn patterns from data instead of being programmed with every possible rule. Supervised learning uses labeled data; unsupervised learning looks for patterns in data without labels, and reinforcement learning learns through rewards and penalties.

Specific Details

For this week's project, I worked with the Wine dataset. The dataset contains different measurements of wine, such as alcohol, malic acid, flavanoids, color intensity, and proline. I used these features to build a classification model that predicts the type of wine. I also experimented with different combinations of three features to see how changing the features affected the model's accuracy.

3. Personal Reflection

Thoughts and Feelings

At first, the different types of machine learning seemed a little confusing because they each have different purposes. After working with the dataset and running the models, I started to understand the differences better. Seeing the model make predictions from the data made the

concepts easier for me to understand.

Analysis and Interpretation

One thing I learned is that the features selected for a model can make a difference in its performance. When I tried different features from the Wine dataset, I could compare the accuracy and see how the choice of data affected the results. This showed me that building a machine learning model is not only about choosing an algorithm. The data and features are also very important.

Connections to Theoretical Knowledge

This project connected directly to what I learned about supervised learning. Since the Wine dataset already has known wine classes, the model can learn from those labeled examples and use them to predict the class of new data. This helped me understand the difference between supervised learning and unsupervised learning.

4. Discussion of Improvements and Learning

Personal Growth

This project helped me become more comfortable working with machine learning data. I learned that it is important to experiment instead of assuming that one model or one group of features will always produce the best results.

Skills Developed

I developed skills in preparing a dataset, selecting features, splitting data into training and testing sets, training a Logistic Regression model, and evaluating its accuracy. I also became more comfortable using Python and libraries such as pandas and scikit-learn.

Future Application

I can apply what I learned to future projects involving classification and prediction. For example,

machine learning could be used in healthcare to predict whether a patient may be at risk for a certain condition based on medical information. I can also use feature selection and model comparisons in future school projects to improve my results.

5. Conclusion

Overall, this week helped me better understand Supervised, Unsupervised, and Reinforcement Learning and how they learn from data in different ways. The project gave me practical experience with supervised learning and showed me how important feature selection is. I also learned that improving a machine learning model requires testing different approaches and analyzing the results. This experience gave me a better foundation for working with machine learning in future projects

Jonathon Wilturner Reflection and Analysis

My Understanding of Machine Learning Types

Supervised Learning: This is when you teach the model using data that already has the right answers attached. You show it examples with the input and the correct output, and it learns to find the pattern between them so it can predict the answer on new data later. The wine classification we did in this lab is a good example, we already knew which wine class each sample belonged to.

Unsupervised Learning: This is when you give the model data with no answers attached at all, and it has to find patterns or groupings on its own. Nobody tells it what the "right" groups are, it just looks for similarities in the data by itself.

Reinforcement Learning:

This is learning by doing, not by being told the answer upfront. The model takes actions, gets a reward or a penalty depending on how it did, and adjusts based on that feedback over time. It's more like learning through trial and error than learning from a fixed answer key.

My Analysis of the Wine Classification Project

Best performing model: Logistic Regression, it came in at 88.9% accuracy, beating the Decision Tree's 83.3%.

Why do you think this model performed better? I think Logistic Regression did better here because the relationships between the wine features and the classes were probably smoother and more linear, which is exactly what Logistic Regression is built to handle. The Decision Tree had a max depth of 3, which limits how complex of a pattern it can actually learn, so it may have been too simple to capture everything going on in the data.

What would you try next to improve performance: [Swap in your real result here once you run

Step 1 — for example: "In Task 1, I tried [your three features] and got [your accuracy]%, which was [higher/lower] than the original model. Next I'd try combining more features together instead of just three, or scaling the data before training, since some features like proline have much bigger numbers than others like hue, which could be throwing the model off."]

Real-World Application Ideas

Industry of Interest: Logistics and shipping.

ML Problem: Predicting which delivery routes are likely to run late based on traffic, weather, and driver history.

Type of ML: Supervised Learning, since you'd be training on past deliveries where you already know if they were late or on time.

Data Needed: Historical delivery times, traffic data, weather conditions, distance, and driver performance history.

Key Learnings

Most important concept learned: Seeing the difference between supervised and unsupervised learning in a real hands-on way made it click more than just reading the definitions. Having labels versus not having labels really does change the whole workflow.

Most challenging part: Understanding why one model beat the other wasn't obvious right away, accuracy numbers alone don't tell you the full story, you have to think about what the model is actually doing underneath.

Questions for further exploration: I'd want to know how much scaling the data would actually change these results, and whether a deeper Decision Tree would start beating Logistic Regression if I let it grow more

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